What we found and what is next

Preliminary findings • Controlled, noise-free regression • Full-batch GD vs L-BFGS

MAIN FINDING

As κ grows, GD needs far more updates.

746

GD updates

5

L-BFGS iterations

At κ = 100 • Same starting weights and gradient-norm threshold

NEXT STEPS

1. **Repeat with new starting points**
2. **Measure gradient calls and runtime**
3. **Try mini-batch SGD or feature scaling**

KEY TAKEAWAY Curvature information helps L-BFGS converge in fewer iterations on narrow loss valleys.

Preliminary results • All six runs met $||\text{gradient}|| < 10^{-6}$ • Fewer iterations ≠ lower total computational cost